

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
 CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – CASE STUDY ANALYSIS

Course Code: DSA05	Course Title: Query Processing for Data Science With Real Time Applications	CO Assessed: CO1
Assessment: Tool 3 – Case Study Analysis	Weightage / Total Marks: 20 Marks	Assessment Mode: Individual Submission
Student Name: [Your Name]	Reg. No.: [Your Reg. No.]	Section / Batch: [Your Section]
Date of Submission: [Date]	Faculty Incharge: Dr.R.Mahesh Muthulakshmi	

CO1 Statement: Explain the fundamentals of data wrangling, its importance, processes, and tools. Understand different data formats such as CSV, JSON, and XML along with machine-readable data. (BL4, BL5 and BL6)

Code of Conduct

I, Edrin J S(192524041), certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: edrinjs

Case Study 1: Student Enrollment Data Integration

Background

A university collects student information from three different departments — Computer Science, Electronics, and Mechanical Engineering — each maintaining records independently in a different data format: the Computer Science department stores data in CSV files, the Electronics department stores data in JSON format, and the Mechanical department stores data in XML format. When the university administration attempts to combine all student records into a central database for academic analysis, several issues are discovered:

- Student names are entered differently (e.g., Raj Kumar, R. Kumar, Rajkumar).
- Some students appear multiple times because they have enrolled in interdisciplinary courses.
- Several records have missing email addresses and phone numbers.
- Date formats differ across departments.

Problem Statement

The university wants to create a unified student information system that can generate accurate reports on student enrollment, attendance, and performance.

Student Tasks

1. Analyze the data quality issues present in the datasets.
2. Categorize the issues into duplicates, missing values, and format inconsistencies.
3. Propose a complete data wrangling process for integrating the datasets.
4. Recommend suitable tools and techniques for data integration.
5. Explain the benefits of clean and integrated student data.

Task 1: Analysis of Data Quality Issues

Before any integration can happen, the three source files were reviewed side by side to identify concrete symptoms of poor data quality. Representative sample records from each department are shown below.

Computer Science Dept – students.csv	Electronics Dept – students.json	Mechanical Dept – students.xml
<pre>RollNo,Name,Email,Phone,DOB CS101,Raj Kumar,raj.k@univ.edu,9876543210,12-05-2003 CS102,R. Kumar,,9876543210,12/05/2003 CS103,Anitha S,anitha.s@univ.edu,,04-11-2002 CS104,Rajkumar,raj.k@univ.edu,9876543210,2003-05-12 CS105,Vivek Rao,vivek.rao@univ.edu,9123456780,21-01-2003</pre>	<pre>{ { "roll_no": "EC201", "name": "Priya Menon", "email": "priya.m@univ.edu", "phone": "9988776655", "dob": "15-08-2003" }, { "roll_no": "EC202", "name": "Raj Kumar", "email": null, "phone": "9876543210", "dob": "2003-05-12" } }</pre>	<pre><students> <student> <roll_no>ME301</roll_no> <name>Rajkumar</name> <email>raj.k@univ.edu</email> <phone></phone> <dob>12.05.2003</dob> </student> <student> <roll_no>ME302</roll_no> <name>Karthik.N</name> <email>karthik.n@univ.edu</email> <phone>9090909090</phone> <dob>03-09-2002</dob> </student> </students></pre>

Fig. 1 — Sample raw records from the three department source files (CSV, JSON, XML).

Comparing these samples surfaces four distinct problems:

- Inconsistent name representation: the same student ("Raj Kumar") appears as "Raj Kumar", "R. Kumar", and "Rajkumar" across CS, EC, and ME records respectively.
- Duplicate identities across departments: interdisciplinary students (e.g., a Computer Science student taking an Electronics elective) are captured once per department, creating 2–3 rows for one real person, each with a different Roll No.
- Missing contact fields: several rows have empty Email or Phone values (e.g., CS102, ME301, EC202), which would break any email/SMS-based communication workflow.
- Inconsistent date formats: DOB appears as DD-MM-YYYY ("12-05-2003"), DD/MM/YYYY ("12/05/2003"), YYYY-MM-DD ("2003-05-12") and even DD.MM.YYYY ("12.05.2003") — four different conventions for the same underlying value.
- Schema mismatch: the CSV, JSON, and XML sources use different field names and nesting (flat columns vs. nested JSON objects vs. XML tags), so they cannot be merged until they are mapped to one common schema.

Task 2: Classification of Issues

The issues identified in Task 1 fall cleanly into three standard data-quality categories used in data wrangling:

Category	Evidence in the Data	Why It Matters
Duplicates	Same student recorded under CS, EC, and ME with different Roll Nos (Raj Kumar / R. Kumar / Rajkumar)	Inflates enrollment counts and skews attendance/performance analytics

Missing Values	Blank Email/Phone fields for CS102, CS103, ME301, EC202	Breaks downstream communication and prevents record matching via contact info
Format Inconsistencies	DOB in 4 different formats; name capitalization/punctuation varies; field names differ across CSV/JSON/XML	Prevents direct joins/sorting and causes incorrect age or seniority calculations

Task 3: Proposed Data Wrangling Process

A seven-stage wrangling pipeline is proposed to take the three raw sources to one clean, integrated Student Information System (SIS) table.

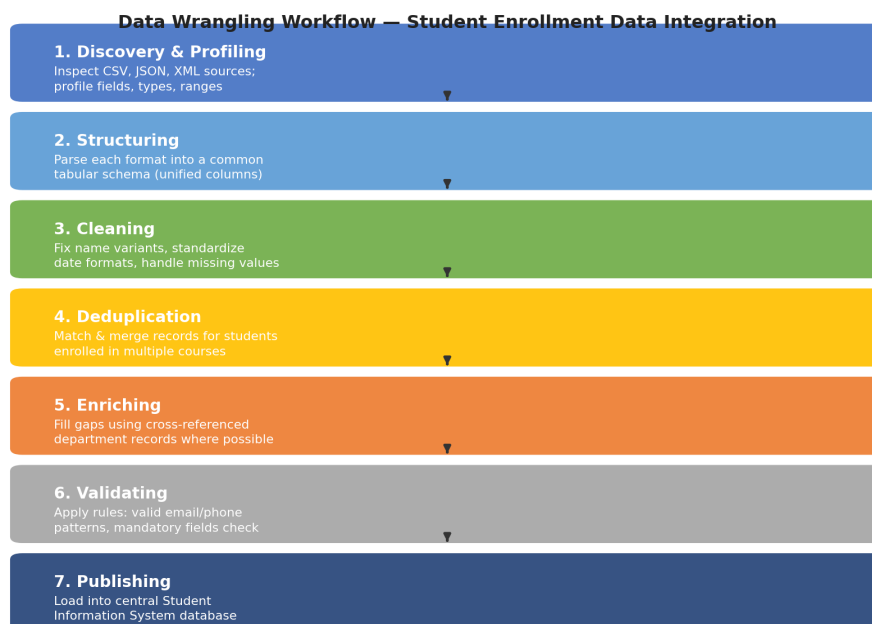


Fig. 2 — End-to-end data wrangling workflow for the enrollment integration.

Stage details

- **Discovery & Profiling:** Load each source and profile column types, null counts, and value ranges to size the problem before touching the data.
- **Structuring:** Parse CSV rows, JSON objects, and XML elements into one common tabular schema (Student ID, Name, Email, Phone, DOB, Department) so the three sources become directly comparable.
- **Cleaning:** Trim whitespace, standardize capitalization, normalize all DOB values to ISO 8601 (YYYY-MM-DD), and apply name-normalization rules (expand initials, remove stray punctuation).
- **Deduplication:** Use fuzzy string matching (e.g., Levenshtein distance / phonetic matching) on normalized names combined with matching DOB or phone number to detect that “Raj Kumar”, “R. Kumar”, and “Rajkumar” are one person, then merge their three records into a single golden record.
- **Enriching:** Where one department has a value another is missing (e.g., CS has the email that ME is missing for the same merged student), fill the gap using the cross-referenced record instead of leaving it blank.
- **Validating:** Apply rule-based checks — valid email pattern, 10-digit phone number, DOB within a plausible age range, mandatory fields present — flagging any record that still fails after cleaning for manual review.

- Publishing: Load the final, validated table into the central relational database that powers enrollment, attendance, and performance reporting.

Task 4: Recommended Tools and Techniques

Tool / Technique	Wrangling Stage	Justification
Python (pandas, json, xml.etree)	Structuring, Cleaning, Validation	Reads CSV/JSON/XML natively, lets each source be converted into a common DataFrame schema, and scripts are re-runnable whenever new department extracts arrive.
OpenRefine	Cleaning, Deduplication	Purpose-built for clustering near-duplicate text values (name variants) using key-collision and n-gram fingerprint algorithms — ideal for “Raj Kumar / R. Kumar / Rajkumar”.
fuzzywuzzy / RapidFuzz (Python library)	Deduplication	Provides Levenshtein-based similarity scoring to automatically flag likely duplicate student names across departments for merge review.
SQL / Relational Database (MySQL, PostgreSQL)	Integration, Publishing	Central store with primary/foreign keys and constraints enforces the single-Student-ID model and supports fast reporting queries across enrollment, attendance and performance.
ETL Orchestration (e.g., Apache Airflow)	Whole Pipeline	Schedules and automates the extract–clean–merge–load steps every semester so the SIS refreshes reliably without manual re-work.

Task 5: Benefits of Clean and Integrated Student Data

The impact of applying the above pipeline can be seen directly by comparing a cluster of raw, unmerged records against the single clean record it produces.

BEFORE — Raw, Unmerged Records

RollNo	Name	Email	Phone	DOB
CS101	Raj Kumar	raj.k@univ.edu	9876543210	12-05-2003
CS102	R. Kumar		9876543210	12/05/2003
ME301	Rajkumar	raj.k@univ.edu		12.05.2003
EC202	Raj Kumar		9876543210	2003-05-12

AFTER — Cleaned & Deduplicated Record

StudentID	Full Name	Email	Phone	DOB (ISO)
S00045	Raj Kumar	raj.k@univ.edu	9876543210	2003-05-12

Fig. 3 — Four raw department entries for one real student merged into a single validated record.

- Accurate reporting: With one golden record per student, enrollment counts, attendance percentages, and performance dashboards reflect real headcounts instead of inflated, duplicate-driven numbers.
- Reliable communication: Standardized, complete email and phone fields mean notices, timetables, and results reach every student without silent delivery failures.

- Cross-department analytics: A common schema and normalized dates let administrators safely join enrollment data with attendance and performance systems for cohort-level analysis.
- Reduced manual rework: Automating the wrangling pipeline removes the need for staff to manually reconcile spreadsheets every semester, saving time and cutting human error.
- Better decision-making: Clean, trustworthy data supports accreditation reporting, resource planning, and early identification of at-risk students — decisions that are only as good as the data behind them.
- Scalability: A repeatable, tool-based process (rather than one-off manual fixes) scales smoothly as the university adds departments or student volume grows.

Conclusion

The Student Enrollment Data Integration case demonstrates why data wrangling is a prerequisite for any centralized information system. The core issues — duplicate identities, missing contact details, and inconsistent formats — are common wherever independent departments capture data in different formats (CSV, JSON, XML). A disciplined, staged wrangling process (discover, structure, clean, deduplicate, enrich, validate, publish), backed by the right mix of scripting (Python), specialized cleaning tools (OpenRefine, fuzzy matching), and a well-modeled relational database, converts fragmented departmental records into one trustworthy, analysis-ready student dataset — directly fulfilling CO1's goal of understanding data wrangling fundamentals, formats, and processes.